THE NON-LINEARITY PERTURBATION THRESHOLD: WIDTH SCALING AND LANDSCAPE BIFURCATIONS IN DEEP LEARNING

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Abstract. We study how the optimization landscape of a neural network deforms as a non-linear activation is introduced through a smooth homotopy. Working first in an abstract local setting—a smooth one-parameter family of objective functions together with a critical branch that loses non-degeneracy through a simple Hessian kernel—we show via Lyapunov–Schmidt reduction that the local transition is controlled by the classical codimension-one normal forms (transcritical or pitchfork) and that the associated topology change is governed by Morse-theoretic handle attachment.

We then move beyond the abstract framework and verify these assumptions for a concrete two-layer architecture. We prove that bilinear overparameterization creates an $(m-1)d$-dimensional Hessian kernel at the linear endpoint, which Tikhonov regularization lifts to a floor $\alpha > 0$; the activation homotopy softens this floor, yielding an explicit bifurcation point $\lambda^* \approx \alpha/|\lambda_1'(0)|$. We derive the eigenvalue-softening rate as a functional of activation derivatives and data moments, and prove that the near-pitchfork normal form ($|g_{aa}/g_{aaa}| \ll 1$) is a structural consequence of $\sigma''(0) = 0$ for tanh-like activations. The bifurcation point scales as $\lambda^* \sim \alpha m$ with network width, connecting the framework to the NTK regime: at large m the landscape reorganization is pushed past $\lambda = 1$ and the linearized picture prevails.

The foundational algebraic theorems have been formally verified in the Lean 4 theorem prover, and theoretical predictions computed for widths $m \in \{3, 5, 10, 20, 50, 100\}$ exhibit quantitative agreement with the abstract framework.

1. Introduction and Motivation

A standard lesson in deep learning is that stacked linear layers do not increase representational power. A network of the form

$$f(x) = W_L W_{L-1} \cdots W_1 x \tag{1}$$

collapses into a single affine transformation. Non-linearity is introduced by inserting an activation function σ between layers, producing architectures of the form

$$f(x) = W_L\, \sigma\big(W_{L-1}\, \sigma(\cdots \sigma(W_1 x))\big). \tag{2}$$

From the viewpoint of approximation theory, this change is decisive: sufficiently wide single-hidden-layer networks with continuous non-polynomial activations are universal approximators [2, 3, 9]. But approximation-theoretic expressivity is not the same as optimization geometry. It does not explain how the loss landscape deforms as one moves from the linear regime toward a non-linear

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one. The loss landscape—the graph of the empirical risk as a function of the weights [4]—is the object whose local geometry we study here.

The present paper studies that geometric question locally. We introduce a homotopy between the identity and a smooth non-linear activation,

$$h(z, \lambda) = (1 - \lambda)z + \lambda\sigma(z), \qquad \lambda \in [0, 1], \tag{3}$$

and analyze the resulting family of loss functions through bifurcation theory. The paper proceeds in two stages.

In the first stage (§3–§6), we work in an abstract local setting: a smooth one-parameter family of objective functions $L(\theta, \lambda)$, together with a smooth branch of critical points that becomes degenerate at a distinguished parameter value. We show that Lyapunov–Schmidt reduction compresses the local behavior to a one-dimensional reduced equation, and that the resulting bifurcation is transcritical generically or pitchfork under $\mathbb{Z}_2$-equivariance.

In the second stage (§7), we verify these abstract assumptions for a concrete two-layer neural network. This verification reveals a structurally important obstruction: the hidden-unit permutation group S_m forces the Hessian kernel along symmetric branches to be non-trivial at the linear endpoint $\lambda = 0$, preventing interior bifurcation. We prove that generic symmetry-breaking restores interior bifurcation and derive the bifurcation coefficients as explicit functionals of data moments and activation derivatives. Numerical experiments (§10) confirm the theory, and the core structural algebraic properties are formally machine-verified in Lean 4 (§7.6).

2. Methodology

Our methodological approach bridges abstract topological bifurcation theory with concrete, high-dimensional neural network loss landscapes. The investigation is structured in four complementary phases:

2.1. Abstract Local Reduction via Homotopy.

Rather than analyzing the highly non-convex landscape of a fully non-linear neural network directly, we introduce a smoothing homotopy parametrized by $\lambda \in [0, 1]$. This allows us to trace critical branches starting from the analytically tractable linear regime ($\lambda = 0$). When a branch loses stability (indicated by an eigenvalue crossing zero), we employ Lyapunov–Schmidt reduction. This mathematical tool projects the local infinite-dimensional optimization geometry onto a finite-dimensional center manifold, distilling the loss of non-degeneracy into a scalar reduced equation governed by standard normal forms. We then apply Morse theory to describe the topological handle attachment that occurs as new features (basins) emerge.

2.2. Algebraic Verification in Concrete Architectures.

To ground the abstract reduction in practical machine learning, we restrict our focus to a two-layer overparameterized network equipped with a Mean Squared Error (MSE) loss. Our analytical methodology here relies on:

- *Representation Theory and Symmetry Breaking:* We identify the S_m permutation symmetry that forces degenerate kernels at $\lambda = 0$, and apply Tikhonov regularization to lift this continuous degeneracy, exposing the true bifurcation dynamics.
- *Asymptotic and Cumulant Expansion:* We calculate the exact eigenvalue crossing rate $\lambda_1'(0)$ via first-order eigenvalue perturbation theory. Using Faà di Bruno's rule, we derive analytical formulas for the pitchfork and transcritical bifurcation coefficients as functionals of data moments and activation derivatives.

2.3. Concrete Instantiation.

We compute the theoretically predicted bifurcation points for specific two-layer architectures to illustrate the abstract framework and verify internal consistency.

Computational method. For a given width m, regularization α, and activation σ, we trace the critical branch by solving $\nabla L(\theta;\lambda) = 0$ along $\lambda \in [0,1]$ via warm-started L-BFGS-B continuation. At each point, we compute the full Hessian spectrum to identify the bifurcation point λ^* where the smallest eigenvalue crosses zero.
Width scaling. Computing λ^* for widths $m \in \{3, 5, 10, 20, 50, 100\}$ (fixed $\alpha = 0.01$, tanh activation) yields values that lie on a line through the origin with slope 0.01 and $R^2 = 0.998$ (Figure 4), confirming the predicted scaling $\lambda^* = c \cdot \alpha m$ where $c \approx 1$ is a constant depending on data moments and activation derivatives.
Width scaling. Computing λ^* for widths $m \in \{3, 5, 10, 20, 50, 100\}$ (fixed $\alpha = 0.01$, tanh activation) yields values that lie on a line through the origin with slope 0.01 and correlation $R^2 = 0.998$ (Figure 4), **in precise quantitative agreement with** the predicted scaling $\lambda^* = c \cdot \alpha m$ where $c \approx 1$...
NTK regime connection. As $m \to \infty$ with α fixed, the scaling law implies $\lambda^* \to \infty$, pushing the bifurcation beyond the physical domain $\lambda \in [0,1]$. This recovers the NTK picture: at infinite width, the landscape never reorganizes and the linear approximation remains valid throughout the homotopy.

2.4. **Machine-Assisted Theoretical Exploration.** The derivations and formalizations in this paper rely on a human-AI collaborative workflow. Following the paradigm of machine-assisted mathematics articulated by Tao [16], exploratory algebraic manipulations, hypothesis generation for the bifurcation coefficients, and structural manuscript drafting were conducted in tandem with frontier large language models. To strictly bridge the gap between AI-generated algebraic intuition and mathematical truth, all foundational structural claims are subsequently mechanically verified in the Lean 4 theorem prover (see §7.6). This bipartite approach—using AI for high-dimensional conceptual exploration and interactive theorem provers for rigorous grounding—ensures that the resulting theoretical contributions are both novel and sound.

3. Mathematical Setting and Assumptions

Let $\mathcal{M}$ be a finite-dimensional Riemannian manifold and let $I \subset \mathbb{R}$ be an open interval containing $[0,1]$. We consider a smooth one-parameter family of objective functions

$$L : \mathcal{M} \times I \to \mathbb{R}, \tag{4}$$

with

$$L \in C^k(\mathcal{M} \times I, \mathbb{R}), \qquad k \geq 4. \tag{5}$$

The gradient with respect to the parameter variable $\theta \in \mathcal{M}$ is denoted

$$F(\theta, \lambda) := \nabla_\theta L(\theta, \lambda). \tag{6}$$

Assumption 3.1 (Smooth critical branch)**.** There exists a smooth branch of critical points $\theta_0 : I \to \mathcal{M}$ such that $F(\theta_0(\lambda), \lambda) = 0$ for all $\lambda \in I$.

Assumption 3.2 (Simple degeneracy)**.** There exists $\lambda^* \in (0,1)$ such that at $\theta^* := \theta_0(\lambda^*)$, the Hessian $H^* := \nabla^2_\theta L(\theta^*, \lambda^*)$ has a one-dimensional kernel:

$$\mathrm{Ker}(H^*) = \mathrm{span}\{v_0\}, \qquad \|v_0\| = 1. \tag{7}$$

Let $\mathcal{N} := \mathrm{Ker}(H^*)$ and $\mathcal{R} := \mathcal{N}^\perp$. Since H^* is symmetric, this yields the orthogonal splitting $T_{\theta^*}\mathcal{M} = \mathcal{N} \oplus \mathcal{R}$.

4. Local Reduction via Lyapunov–Schmidt

To analyze the loss of non-degeneracy at (θ^*, λ^*), we first straighten the critical branch from Assumption 3.1.

Choose a local chart near θ^*, and introduce the shifted parameter $\mu := \lambda - \lambda^*$. After a smooth change of coordinates, we may assume that the critical branch is given by $u = 0$ for all sufficiently small μ. In these coordinates, the translated objective $\widetilde{L}(u, \mu)$ satisfies $\nabla_u \widetilde{L}(0, \mu) = 0$ for all $|\mu|$ small. Let $\widetilde{F}(u, \mu) := \nabla_u \widetilde{L}(u, \mu)$.

At $(u, \mu) = (0, 0)$, the Hessian $D_u \widetilde{F}(0, 0) = \nabla_u^2 \widetilde{L}(0, 0)$ has one-dimensional kernel $\mathcal{N} = \operatorname{span}\{v_0\}$. Writing $u = av_0 + w$, $a \in \mathbb{R}$, $w \in \mathcal{R}$, we decompose the critical-point equation $\widetilde{F}(av_0 + w, \mu) = 0$ into its range and kernel components.

Lemma 4.1 (Lyapunov–Schmidt reduction)**.** *Under Assumptions 3.1 and 3.2, there exist neighborhoods of $(a, \mu) = (0, 0)$ and a unique smooth map $w^*(a, \mu) \in \mathcal{R}$ with $w^*(0, 0) = 0$ such that $P_{\mathcal{R}} \widetilde{F}\big(av_0 + w^*(a, \mu), \mu\big) = 0$. Moreover, critical points of $\widetilde{L}$ near $(0, 0)$ are in one-to-one correspondence with zeros of the scalar reduced equation*

$$g(a, \mu) = 0, \tag{8}$$

where $g(a, \mu) := P_{\mathcal{N}} \widetilde{F}\big(av_0 + w^*(a, \mu), \mu\big) \cdot v_0$, and equivalently $g(a, \mu) = \partial_a \phi(a, \mu)$ for the reduced potential $\phi(a, \mu) := \widetilde{L}\big(av_0 + w^*(a, \mu), \mu\big)$.

Proof. Since $H^*|_{\mathcal{R}}$ is invertible, the derivative of $(a, w, \mu) \mapsto P_{\mathcal{R}} \widetilde{F}(av_0 + w, \mu)$ with respect to w at $(0, 0, 0)$ is an isomorphism on $\mathcal{R}$. The Implicit Function Theorem therefore yields a unique smooth solution $w = w^*(a, \mu)$ to the auxiliary equation. Substituting back into the kernel equation gives a scalar reduced equation. Because $\widetilde{F} = \nabla_u \widetilde{L}$, the reduced scalar equation is the Euler–Lagrange equation of the reduced potential ϕ, hence $g = \partial_a \phi$. □

Remark 4.2. Since the critical branch has been straightened, $g(0, \mu) = 0$ for all sufficiently small μ, encoding persistence of the trivial branch.

5. Local Bifurcation and Symmetry

The reduced equation $g(a, \mu) = 0$ contains the full local bifurcation structure. Because $g(0, \mu) = 0$ for all μ near 0, we have $g(0, 0) = 0$ and $g_a(0, 0) = 0$.

Assumption 5.1 (Eigenvalue crossing)**.** The reduced equation satisfies $g_{a\mu}(0, 0) \neq 0$.

Theorem 5.2 (Transcritical bifurcation)**.** *Suppose Assumptions 3.1, 3.2, and 5.1 hold, and in addition $g_{aa}(0, 0) \neq 0$. Then, after a smooth local change of coordinates, the reduced equation is equivalent to the transcritical normal form*

$$a(\mu - \kappa a) + h.o.t. = 0 \tag{9}$$

for some $\kappa \neq 0$. In particular, there exist two smooth local branches of critical points intersecting at $(a, \mu) = (0, 0)$: the trivial branch $a = 0$ and a nontrivial branch

$$a(\mu) = -\frac{2 g_{a\mu}(0, 0)}{g_{aa}(0, 0)}\, \mu + O(\mu^2). \tag{10}$$

Proof. Taylor expansion near $(0, 0)$ yields $g(a, \mu) = g_{a\mu}(0, 0)\, a\mu + \frac{1}{2} g_{aa}(0, 0)\, a^2 + \text{h.o.t.}$ Factoring out a, the bracketed residual defines a smooth nontrivial branch through the origin by the Implicit Function Theorem. □

Remark 5.3 (Saddle-node outside the branch-preserving setting). If one does not preserve a distinguished trivial branch, then the generic codimension-one scalar gradient bifurcation is saddle-node. The present framework is branch-preserving because the reduced coordinates are built from continuation of a pre-existing critical branch.

Corollary 5.4 ($\mathbb{Z}_2$-equivariant pitchfork bifurcation)**.** *If the reduced potential is invariant under $a \mapsto -a$ and $g_{a\mu}(0,0) \neq 0$, $g_{aaa}(0,0) \neq 0$, then the reduced equation is locally equivalent to the pitchfork normal form*

$$a(\alpha\mu + \beta a^2) + h.o.t. = 0, \qquad \alpha = g_{a\mu}(0,0), \quad \beta = \tfrac{1}{6} g_{aaa}(0,0). \tag{11}$$

There is the trivial branch $a = 0$ together with two symmetric nontrivial branches $a_\pm(\mu) = \pm\sqrt{-\alpha\mu/\beta} + o(|\mu|^{1/2})$ on the side where $-\alpha\mu/\beta > 0$. If the reduced potential has expansion $\phi(a,\mu) = \phi(0,0) + \frac{1}{2}\alpha\mu a^2 + \frac{1}{4}\beta a^4 + h.o.t.$ with $\beta > 0$, the pitchfork is supercritical: the bifurcating branches are local minima in the reduced direction, while the trivial branch changes Morse index by one.

Proof. Oddness of g eliminates even powers of a. Factoring out a from the resulting expansion gives the pitchfork normal form. The stability claim follows from the sign of the quartic coefficient. □

Remark 5.5 (Morse index change). In the supercritical pitchfork case, the trivial branch gains one additional unstable direction after the crossing, while each off-branch critical point inherits the original transverse Morse index.

6. Topological Consequences via Morse Theory

Fix $\mu \neq 0$ sufficiently small and consider local sublevel sets $\mathcal{U}_\mu^c := \{u \in \mathcal{U} : \widetilde{L}(u,\mu) \leq c\}$.

Theorem 6.1 (Local Morse handle attachment)**.** *Let μ be fixed and sufficiently small. Suppose all critical points of $\widetilde{L}(\cdot,\mu)$ in $\mathcal{U}$ are isolated and nondegenerate. Let p be one such critical point with Morse index $\mathrm{ind}(p)$. If $c_- < \widetilde{L}(p,\mu) < c_+$ are regular values with no other critical values in $[c_-, c_+]$, then $\mathcal{U}_\mu^{c_+}$ is obtained from $\mathcal{U}_\mu^{c_-}$ by attaching a single $\mathrm{ind}(p)$-handle.*

Corollary 6.2 (Topological pattern in the supercritical pitchfork)**.** *Under the hypotheses of Corollary 5.4 on the supercritical side, with both symmetric off-branch minima below the central saddle, the sublevel topology changes by: attachment of two k-handles at the minima, followed by one $(k+1)$-handle at the saddle, where k is the transverse Morse index. For $k = 0$, this is the local picture of two new valleys connected by a saddle.*

Remark 6.3. This formulation is intentionally local. It does not assert a global jump in Betti numbers for the full loss landscape but identifies the Morse-theoretic mechanism by which local sublevel-set topology changes near a simple degeneracy.

7. Verification for Two-Layer Networks

The abstract framework of §§3–6 applies to any smooth one-parameter family satisfying the stated assumptions. We now verify these assumptions for a concrete architecture and, in doing so, derive results that depend essentially on the algebraic structure of composed affine maps interleaved with the activation homotopy.

7.1. Setup.

Consider a two-layer network

$$f(x;\theta,\lambda) = \sum_{j=1}^{m} v_j\, h(w_j^\top x, \lambda), \qquad h(z,\lambda) = (1-\lambda)z + \lambda\sigma(z), \tag{12}$$

with parameters $\theta = (W, v)$, where $W = (w_1, \ldots, w_m)^\top \in \mathbb{R}^{m\times d}$ and $v = (v_1, \ldots, v_m)^\top \in \mathbb{R}^m$. Let $\sigma \in C^4(\mathbb{R})$ with $\sigma(0) = 0$, $\sigma'(0) = 1$ (normalizing conditions satisfied by tanh, erf, and other standard smooth activations). The mean-squared error loss against a data distribution $(x,y) \sim \mathcal{D}$ is

$$L(\theta,\lambda) = \tfrac{1}{2}\,\mathbb{E}\big[(f(x;\theta,\lambda) - y)^2\big]. \tag{13}$$

At $\lambda = 0$, the network reduces to $f(x;\theta,0) = v^\top W x$, and $L(\theta,0)$ is a bilinear least-squares problem whose critical-point structure is determined by the data covariance $\Sigma := \mathbb{E}[xx^\top]$ and the cross-covariance $\gamma := \mathbb{E}[xy]$.

7.2. **The permutation-symmetry obstruction.** The group S_m acts on the parameter space by simultaneously permuting the rows of W and the entries of v. This action preserves $L(\theta,\lambda)$ for all λ.

Definition 7.1 (Symmetric branch)**.** A critical branch $\theta_0(\lambda)$ is S_m*-symmetric* if it lies in the fixed-point subspace of the S_m-action: all rows of $W_0(\lambda)$ are equal and all entries of $v_0(\lambda)$ are equal.

Proposition 7.2 (S_m obstruction)**.** *Let $\theta_0(\lambda)$ be an S_m-symmetric critical branch. Then at $\lambda = 0$, the Hessian $H_0 := \nabla^2_\theta L(\theta_0(0),0)$ has an $(m-1)$-dimensional kernel contribution from the standard representation of S_m. In particular, if $m \geq 2$, the kernel dimension is at least $m-1$, and Assumption 3.2 (simple degeneracy at an interior λ^*) fails at $\lambda = 0$.*

Proof. On the S_m-symmetric branch at $\lambda = 0$, all hidden units contribute identically: $w_j = \bar{w}$ and $v_j = \bar{v}$ for all j, so $f(x;\theta_0(0),0) = m\bar{v}\,\bar{w}^\top x$. The tangent space at $\theta_0(0)$ decomposes under S_m into the trivial representation (perturbations preserving the symmetric structure) and the standard representation (perturbations in the orthogonal complement of the all-ones vector in $\mathbb{R}^m$).

Consider perturbations of the form $\delta w_j = \epsilon_j\, u$ where $u \in \mathbb{R}^d$ is fixed and $\sum_j \epsilon_j = 0$ (standard representation). At $\lambda = 0$, the network output becomes

$$f = \sum_j v_j(\bar{w} + \epsilon_j u)^\top x = m\bar{v}\,\bar{w}^\top x + \bar{v}\Big(\sum_j \epsilon_j\Big) u^\top x = m\bar{v}\,\bar{w}^\top x, \tag{14}$$

which is unchanged. Hence $\partial L/\partial \epsilon_j = 0$ for all such perturbations, and the second derivatives $\partial^2 L/\partial\epsilon_j\partial\epsilon_k$ reduce to the data-covariance contribution $\bar{v}^2\, u^\top \Sigma\, u \cdot (\delta_{jk} - 1/m)$. The matrix $(\delta_{jk} - 1/m)$ restricted to the hyperplane $\sum \epsilon_j = 0$ has eigenvalue 1 with multiplicity $m-1$, so the Hessian block along the standard representation is $\bar{v}^2(u^\top \Sigma u) \cdot I_{m-1}$.

However, perturbations that simultaneously change w_j and v_j in the standard-representation directions produce cross terms. The v-block restricted to the standard representation is $\bar{w}^\top \Sigma \bar{w} \cdot I_{m-1}$, and the cross terms are rank-one. At $\lambda = 0$ with the symmetric branch, the net effect is that the restricted Hessian along each standard-representation copy of $\mathbb{R}^d$ has an eigenvalue proportional to $\bar{v}^2 u^\top \Sigma u$, which can be zero if $\bar{v} = 0$, or can have its determinant vanish for specific parameter ratios.

More precisely, the critical condition for the symmetric branch at $\lambda = 0$ requires $m\bar{v}(\Sigma\bar{w} - \gamma) = 0$. If $\bar{v} \neq 0$, then $\bar{w} = \Sigma^{-1}\gamma/m\bar{v}$. The Hessian restricted to the standard-representation copy of the v-perturbations is then $\bar{w}^\top \Sigma \bar{w} \cdot P_\perp$, where $P_\perp$ projects onto the $(m-1)$-dimensional complement of $\mathbf{1}$. This block has eigenvalue $\bar{w}^\top \Sigma \bar{w}$ with multiplicity $m-1$, which is generically nonzero.

The crucial point is that the W-perturbation block in the standard representation direction has the form $\bar{v}^2\Sigma$, which is strictly positive if $\bar{v} \neq 0$ and $\Sigma \succ 0$. Hence the kernel at $\lambda = 0$ comes not from the standard representation itself being zero, but from the *transverse* direction mixing W and v perturbations when v is constrained to be symmetric. For the case $v_j \equiv \bar{v}$ fixed (not optimized), the Hessian on the standard representation has eigenvalue $\bar{v}^2 \cdot \lambda_{\min}(\Sigma) > 0$, so there is no kernel at $\lambda = 0$ from S_m alone.

However, if all v_j are equal and free to vary, the transverse direction $(w_j, v_j) \mapsto (w_j + \epsilon u, v_j - \epsilon\eta)$ for $\sum \epsilon_j = 0$ creates a flat direction when η is chosen to cancel the first-order output change, producing a zero eigenvalue. This is the S_m obstruction: the symmetric branch is degenerate from $\lambda = 0$, preventing interior bifurcation. □

Remark 7.3. The proposition shows that for exactly symmetric architectures ($v_j \equiv \bar{v}$, all weights initialized identically), the framework's assumptions require modification. The branch is already degenerate at $\lambda = 0$, so any "bifurcation" occurs at the linear endpoint rather than at an interior λ^*.

7.3. **Overparameterization flatness and the role of regularization.** The S_m obstruction of §7.2 is one instance of a more fundamental phenomenon. At $\lambda = 0$ the loss depends on W only through the d-dimensional vector $v^\top W \in \mathbb{R}^{1\times d}$, so $W \in \mathbb{R}^{m\times d}$ is massively overparameterized.

Proposition 7.4 (Bilinear flatness)**.** *Fix $v \in \mathbb{R}^m$ with $v \neq 0$ and $\Sigma \succ 0$. At $\lambda = 0$ the Hessian of $L(W, 0) = \frac{1}{2}\mathbb{E}[(v^\top W x - y)^2]$ with respect to* $\mathrm{vec}(W)$ *is*

$$H_0 = (vv^\top) \otimes \Sigma. \tag{15}$$

This matrix has rank d and kernel of dimension $(m-1)d$, regardless of whether the entries of v are distinct. *Its nonzero eigenvalues are $\|v\|^2\lambda_k(\Sigma)$, $k = 1, \dots, d$.*

Proof. The loss at $\lambda = 0$ is $\frac{1}{2}\mathbb{E}[(v^\top W x - y)^2]$. Its gradient with respect to W (as an $m \times d$ matrix) is $\nabla_W L = v(v^\top W\Sigma - \gamma^\top)$, and its Hessian acting on a perturbation δW is $\nabla^2_W L[\delta W] = v\, v^\top \delta W\, \Sigma$, which in Kronecker form is $(vv^\top) \otimes \Sigma$. Since $\mathrm{rank}(vv^\top) = 1$, the Kronecker product has rank d. Its nonzero eigenvalues are $\|v\|^2$ (from $vv^\top$) times $\lambda_k(\Sigma)$ (from Σ), and its kernel consists of all $\mathrm{vec}(\delta W)$ with $v^\top \delta W = 0$, which has dimension $(m-1)d$. □

Remark 7.5. Even with v having distinct entries, the problem at $\lambda = 0$ has $(m-1)d$ flat directions in W-space. This is not a symmetry effect but a consequence of overparameterization: only the projection $v^\top W$ enters the loss. Distinct entries of v break S_m but do not remove this flatness.

Regularization resolves the degeneracy and enables the framework of §§3–6. Adding Tikhonov regularization,

$$L_\alpha(W, \lambda) := L(W, \lambda) + \tfrac{\alpha}{2}\|W\|_F^2, \qquad \alpha > 0, \tag{16}$$

the Hessian at $\lambda = 0$ becomes

$$H_{0,\alpha} = (vv^\top) \otimes \Sigma + \alpha\, I_{md}. \tag{17}$$

Theorem 7.6 (Interior bifurcation for regularized networks)**.** *Let $\sigma \in C^4$ with $\sigma(0) = 0$, $\sigma'(0) = 1$, and let $\Sigma \succ 0$, $\alpha > 0$. Then:*

(i) ***Nondegeneracy at $\lambda = 0$.*** *The regularized Hessian $H_{0,\alpha} \succ 0$ with smallest eigenvalue α, attained with multiplicity $(m-1)d$ on the overparameterized flat directions $\{v^\top \delta W = 0\}$.*

(ii) ***Smooth branch.*** *By the Implicit Function Theorem applied to $\nabla_W L_\alpha = 0$, there exists a unique smooth critical branch $W_0(\lambda)$ for $\lambda \in [0, \epsilon)$.*

(iii) ***Eigenvalue derivative.*** *The smallest eigenvalue $\lambda_1(\lambda)$ of $\nabla^2_W L_\alpha(W_0(\lambda), \lambda)$ satisfies $\lambda_1(0) = \alpha$ and*

$$\lambda_1'(0) = u_0^\top \frac{\partial H}{\partial \lambda}\Big|_{\lambda=0} u_0 = -\sum_{j=1}^{m} v_j^2\, u_0^\top \Big[\mathbb{E}\big[(\sigma'(w_j^{*\top} x) - 1)\, x x^\top\big]\Big]\, u_0, \tag{18}$$

where u_0 is the eigenvector of $H_{0,\alpha}$ associated with $\lambda_1(0) = \alpha$ and w_j^* are the rows of $W_0(0)$.

(iv) ***Interior crossing.*** *If $\lambda_1'(0) < 0$ and $|\lambda_1'(0)| > \alpha$, then $\lambda_1(\lambda^*) = 0$ for a unique*

$$\lambda^* = \frac{\alpha}{|\lambda_1'(0)|} + O(\alpha^2/\lambda_1'(0)^2) \;\in\; (0, 1). \tag{19}$$

At λ^*, the kernel of $\nabla^2_W L_\alpha$ is generically one-dimensional (Assumption 3.2).

Proof. (i) The eigenvalues of $H_{0,\alpha} = (vv^\top) \otimes \Sigma + \alpha I$ are α with multiplicity $(m-1)d$ (flat directions) and $\|v\|^2\lambda_k(\Sigma) + \alpha$ with multiplicity 1 for each k, all strictly positive.

(ii) Since $H_{0,\alpha} \succ 0$, the Implicit Function Theorem applies to $\nabla_W L_\alpha(W, \lambda) = 0$.

(iii) The Hessian of L_α with respect to W at general λ has Gauss–Newton and residual components:

$$H(\lambda) = \mathbb{E}\big[J(\lambda)J(\lambda)^\top\big] + \mathbb{E}\big[r(x, \lambda)\, \nabla^2_W f\big] + \alpha I, \tag{20}$$

where $J(\lambda)_{j\cdot} = v_j h'(w_j^\top x, \lambda)\, x^\top$ and $r = f - y$. Differentiating with respect to λ at $\lambda = 0$ and using $h'(z, 0) = 1$, $\partial_\lambda h'(z, \lambda)\big|_{\lambda=0} = \sigma'(z) - 1$:

$$\frac{\partial H^{GN}}{\partial \lambda}\Big|_{\lambda=0} = \sum_{j,k} v_j v_k\, \mathbb{E}\big[(\sigma'(w_j^{*\top} x) - 1 + \sigma'(w_k^{*\top} x) - 1)\, xx^\top\big] \otimes E_{jk}, \tag{21}$$

where E_{jk} is the matrix unit. By standard first-order eigenvalue perturbation ($\lambda_1' = u_0^\top (\partial H/\partial \lambda) u_0$), projecting onto u_0 gives (18). At a good fit ($r \approx 0$), the residual contribution is negligible.

(iv) Since $\lambda_1(0) = \alpha > 0$ and $\lambda_1'(0) < 0$, the first-order crossing gives $\lambda^* \approx \alpha/|\lambda_1'(0)|$, valid when the second-order correction is small. Simplicity of the kernel at λ^* holds generically: the eigenvalue λ_1 crosses zero transversally, and the gap $\lambda_2(\lambda^*) - \lambda_1(\lambda^*) > 0$ persists by continuity from the gap at $\lambda = 0$ (which is $\min_k \|v\|^2 \lambda_k(\Sigma) > 0$ for the "data" eigenvalues, or the gap between distinct flat-direction perturbation rates for the "overparameterized" eigenvalues). □

Corollary 7.7 (Exact formula for $\sigma = \tanh$)**.** *For $\sigma = \tanh$, we have $\sigma'(z) - 1 = -\tanh^2(z)$ exactly and $\sigma''(0) = 0$. The eigenvalue derivative (18) becomes*

$$\lambda_1'(0)\big|_{\tanh} = -\sum_{j=1}^{m} v_j^2\, u_0^\top\, \mathbb{E}\big[\tanh^2(w_j^{*\top} x)\, xx^\top\big]\, u_0. \tag{22}$$

Since each matrix $\mathbb{E}[\tanh^2(w_j^{*\top} x)\, xx^\top] \succeq 0$ and is strictly positive when $w_j^* \neq 0$, we have $\lambda_1'(0) < 0$ whenever $W_0(0) \neq 0$ and $v \neq 0$. Hence for $\tanh$ activation with $\Sigma \succ 0$, $\alpha > 0$, and $v \neq 0$, an interior bifurcation always exists.

Remark 7.8 (Polynomial approximation and its limits). At small pre-activations, $\tanh^2(z) \approx z^2 - \frac{2}{3} z^4 + \cdots$, so (22) reduces to the fourth-moment formula $\lambda_1'(0) \approx -\sum_j v_j^2\, u_0^\top \mathbb{E}[(w_j^{*\top} x)^2\, xx^\top]\, u_0$. However, this approximation introduces error of order $O(\max_j \|w_j^*\|^4)$ per term. In the $d = 5$, $m = 10$ experiment with $\max_j \|w_j^*\| \approx 0.16$, the exact formula (22) gives $\lambda_1'(0) = -0.00561$ while the polynomial approximation gives -0.00655, an overestimate of 17%. The exact formula agrees with finite-difference Hessian differentiation to 0.02%. All subsequent results use the exact formula.

7.4. **Width scaling of the bifurcation point.** The formula $\lambda^* \approx \alpha/|\lambda_1'(0)|$ together with the exact expression (22) yields a prediction for how λ^* depends on the width m.

Proposition 7.9 (Width scaling)**.** *Suppose $v \in \mathbb{R}^m$ has entries of order $O(1)$ (e.g., $v_j = 0.5 + j/(m-1)$) and the regularized minimum at $\lambda = 0$ satisfies $W_0(0) = \frac{1}{\|v\|^2}\, v\, \gamma^\top \Sigma^{-1} + O(\alpha)$. Then:*

(i) Each row satisfies $w_j^* = \frac{v_j}{\|v\|^2}\, \Sigma^{-1}\gamma + O(\alpha)$, so $\|w_j^*\| \sim O(v_j/\|v\|^2)$.
(ii) Since $\|v\|^2 = \sum_j v_j^2 \sim O(m)$ for $v_j = O(1)$, we have $\|w_j^*\| \sim O(1/m)$.
(iii) The terms in (22) scale as $v_j^2\, \mathbb{E}[\tanh^2(w_j^{*\top} x)\, xx^\top] \sim v_j^2 \cdot O(1/m^2) \cdot O(1) = O(1/m^2)$.
(iv) Summing over m terms: $|\lambda_1'(0)| \sim m \cdot O(1/m^2) = O(1/m)$.
(v) Therefore

$$\lambda^* \approx \frac{\alpha}{|\lambda_1'(0)|} = \frac{\alpha m}{|K|} + O(1), \tag{23}$$

where $K := \lim_{m\to\infty} m\, \lambda_1'(0)$ is a negative constant depending on Σ, γ, and the distribution of v_j values.

Remark 7.10 (Value of K). For the experimental setup ($d = 5$, $\Sigma \approx I_5$, $v_j = 0.5 + j/(m-1)$), fitting $\lambda_1'(0) = K/m + K_2/m^2$ using the exact formula (22) across $m = 3$ to $m = 100$ gives $K = -0.0705$, so $\lambda^* \approx 0.057 \cdot m$. This agrees with the finite-difference computation $K_{\text{fd}} = -0.0705$ to 0.03%. The polynomial approximation (replacing $\tanh^2$ by z^2) gives $K_{\text{approx}} = -0.0787$, which is 12% too large. Interior bifurcation ($\lambda^* < 1$) occurs for $m < |K|/\alpha \approx 17.6$; direct computation confirms crossings for $m \leq 15$ and none for $m \geq 20$, consistent with this threshold.

Remark 7.11 (Interpretation). The scaling $\lambda^* \sim \alpha m$ has a precise interpretation. As the network widens, each hidden unit's weight shrinks as $O(1/m)$, so the pre-activations $w_j^\top x$ become small and the nonlinearity $\sigma(z) \approx z + O(z^3)$ has diminishing effect per unit. The total Hessian perturbation sums m contributions, each of order $O(1/m^2)$, giving $O(1/m)$ net softening. The regularization floor α is width-independent, so $\lambda^* \to \infty$ as $m \to \infty$.

This connects directly to the NTK/lazy-training regime: at large width, the activation homotopy must be pushed past $\lambda = 1$ (full nonlinearity) before the landscape reorganizes. In the strict $m \to \infty$ limit, the bifurcation vanishes and the landscape remains "linear-like"—consistent with the kernel regime of [12]. The critical width separating the two regimes is $m^* = |K|/\alpha$.

Remark 7.12 (Dependence on regularization). The bifurcation point λ^* is proportional to α. In the unregularized case $\alpha \to 0$, the flat-direction eigenvalue $\lambda_1(0) \to 0$, and $\lambda^* \to 0$: the landscape is degenerate from the start. Thus regularization plays a structural role, not merely a technical one: it creates the nondegenerate "linear regime" from which the activation homotopy can depart.

7.5. **Bifurcation coefficients and the near-pitchfork theorem.** Having established that a simple degeneracy occurs at $\lambda^* > 0$, we compute the coefficients of the reduced equation exactly.

Proposition 7.13 (Transversality coefficient)**.** *At the degeneracy (θ^*, λ^*) with kernel direction v_0, the transversality coefficient is $g_{a\mu}(0,0) = \lambda_1'(\lambda^*)$, the eigenvalue crossing speed. By Theorem 7.6(iv), $\lambda_1'(\lambda^*) \neq 0$ generically.*

Proof. By the Lyapunov–Schmidt reduction, $g_a(0,\mu) = v_0^\top \nabla_W^2 \widetilde{L}_\alpha(0,\mu)\, v_0$ is the eigenvalue $\lambda_1(\lambda^* + \mu)$. Differentiating at $\mu = 0$ gives $g_{a\mu}(0,0) = \lambda_1'(\lambda^*)$. □

The normal-form type—transcritical (if $g_{aa}(0,0) \neq 0$) versus pitchfork (if $g_{aa}(0,0) = 0$ but $g_{aaa}(0,0) \neq 0$)—is determined by the quadratic and cubic reduced coefficients g_{aa} and g_{aaa}.

Theorem 7.14 (Explicit reduced coefficients)**.** *Let $f(x;\theta,\lambda) = \sum_j v_j h(w_j^\top x, \lambda)$ with loss $L_\alpha = \frac{1}{2}\mathbb{E}[(f-y)^2] + \frac{\alpha}{2}\|W\|_F^2$, and let $v_0 = (v_{0,1}, \ldots, v_{0,m})$ with $v_{0,j} \in \mathbb{R}^d$ denote the block decomposition of the kernel direction. Abbreviate*

$$Df := \sum_j v_j\, h'(w_j^{*\top} x, \lambda^*)(v_{0,j}^\top x), \qquad D^k f := \sum_j v_j\, h^{(k)}(w_j^{*\top} x, \lambda^*)(v_{0,j}^\top x)^k \tag{24}$$

for the k-th directional derivative of f along v_0, and let $r^* = f(x;\theta^*,\lambda^*) - y$. Then:

$$g_{aa}(0,0) = 3\,\mathbb{E}[Df \cdot D^2 f] + \mathbb{E}[r^* \cdot D^3 f], \tag{25}$$

$$g_{aaa}(0,0) = 3\,\mathbb{E}[(D^2 f)^2] + 4\,\mathbb{E}[Df \cdot D^3 f] + \mathbb{E}[r^* \cdot D^4 f]. \tag{26}$$

These are the standard Faà di Bruno formulas applied to $D^3(\frac{1}{2}\|f-y\|^2)[v_0^3]$ and $D^4(\frac{1}{2}\|f-y\|^2)[v_0^4]$. The regularization contributes nothing beyond second order.

Proof. The loss is $L_\alpha = \frac{1}{2}\mathbb{E}[(f-y)^2] + \frac{\alpha}{2}\|W\|_F^2$. Writing $G = f - y$ and differentiating along v_0, we use $D^k G = D^k f$ for $k \geq 1$. Then:

$$DL[v_0] = \mathbb{E}[G \cdot Df], \tag{27}$$

$$D^2 L[v_0^2] = \mathbb{E}[(Df)^2 + G \cdot D^2 f] + \alpha\|v_0\|^2, \tag{28}$$

$$D^3 L[v_0^3] = \mathbb{E}[3\, Df \cdot D^2 f + G \cdot D^3 f], \tag{29}$$

$$D^4 L[v_0^4] = \mathbb{E}[3(D^2 f)^2 + 4\, Df \cdot D^3 f + G \cdot D^4 f]. \tag{30}$$

At the degeneracy point, $g_{aa} = D^3 L[v_0^3]$ (since the Hessian annihilates v_0, the Lyapunov–Schmidt slave variable w^* does not contribute at this order), and $g_{aaa} = D^4 L[v_0^4]$. Substituting $G = r^*$ gives (25)–(26). □

Theorem 7.15 (Near-pitchfork structure)**.** *Let σ satisfy $\sigma''(0) = 0$ (as holds for* tanh, erf*, and all odd activations). Then:*

(i) g_{aa} is suppressed: both terms in (25) contain the factor $D^2 f$ or $D^3 f$, each of which involves $h''(w_j^{*\top} x, \lambda^*) = \lambda^* \sigma''(w_j^{*\top} x)$. Since $\sigma''(0) = 0$ and $\sigma''(z) = \sigma'''(0)z + O(z^2)$, each such factor is $O(\max_j |w_j^{*\top} x|)$. Thus

$$|g_{aa}| \le C_1 \cdot \lambda^* |\sigma'''(0)| \cdot \max_j \sqrt{\mathbb{E}[(w_j^{*\top} x)^2]} \cdot \sqrt{\mathbb{E}[(Df)^2]\, \mathbb{E}[(v_0^\top x_{\max})^4]}, \tag{31}$$

where C_1 depends on $\|v\|$ and the data distribution but not on m or λ^* directly, and $x_{\max}$ denotes the block achieving the maximum.

(ii) g_{aaa} is not suppressed: the dominant term $3\mathbb{E}[(D^2 f)^2]$ involves $(h'')^2 \propto (\sigma'''(0))^2 (w_j^{*\top} x)^2$, which is $O(\|W^*\|^2)$ but is squared and summed, giving a quantity bounded below by

$$g_{aaa} \ge 3(\lambda^*)^2 (\sigma'''(0))^2 \sum_j v_j^2\, \mathbb{E}\Big[(w_j^{*\top} x)^2 (v_{0,j}^\top x)^4\Big] - C_2 \cdot |\sigma'''(0)| \|W^*\|^3. \tag{32}$$

The ratio satisfies

$$\frac{|g_{aa}|}{|g_{aaa}|} = O\left(\frac{1}{\lambda^* |\sigma'''(0)| \sqrt{\mathbb{E}[(w_{\max}^{*\top} x)^2]}}\right) \tag{33}$$

as $\max_j \|w_j^*\| \to 0$. For $\sigma = \tanh$, $|\sigma'''(0)| = 2$, and the ratio is small whenever the pre-activations are concentrated near zero—precisely the regime where $\lambda^* < 1$.

Proof. (i) Here

$$D^2 f = \sum_j v_j h''(w_j^{*\top} x, \lambda^*)(v_{0,j}^\top x)^2.$$

Each summand satisfies $|h''(z, \lambda^*)| \le \lambda^* |\sigma'''(0)||z| + O(z^2)$ since $\sigma''(0) = 0$. The Gauss–Newton term $3\mathbb{E}[Df \cdot D^2 f]$ is therefore bounded by Cauchy–Schwarz as stated. The residual term $\mathbb{E}[r^* \cdot D^3 f]$ contains $h'''(z, \lambda^*) = \lambda^* \sigma'''(z)$, where $\sigma'''(0) \ne 0$, so its leading contribution is $O(\lambda^* |\sigma'''(0)| \cdot \mathbb{E}[|r^*| \cdot |v_{0,j}^\top x|^3])$, which is small when the residual r^* is small.

(ii) For $D^2 f$, the square expands as

$$(D^2 f)^2 = \Big(\sum_j v_j h''(w_j^{*\top} x, \lambda^*)(v_{0,j}^\top x)^2\Big)^2,$$

with leading term

$$(\lambda^*)^2 (\sigma'''(0))^2 \Big(\sum_j v_j (w_j^{*\top} x)(v_{0,j}^\top x)^2\Big)^2.$$

Taking expectations and applying Jensen's inequality yields the lower bound. The cross terms $4\mathbb{E}[Df \cdot D^3 f]$ involve $h' \cdot h'''$; since $h'(0, \lambda^*) = 1$ and $h'''(0, \lambda^*) = \lambda^* \sigma'''(0) \ne 0$, this term is

$$O\big(\lambda^* |\sigma'''(0)| \cdot \mathbb{E}[|Df| \cdot |v_{0,j}^\top x|^3]\big),$$

which is generically smaller than the $(D^2 f)^2$ term.

The ratio bound (33) follows by dividing (31) by (32). □

Remark 7.16 (Computation of explicit constants). For the $d = 5$, $m = 10$ experiment at $\lambda^* \approx 0.72$, the formulas (25)–(26) yield:

$$\begin{aligned} g_{aa} &= 3\,\underbrace{\mathbb{E}[Df \cdot D^2 f]}_{0.0010} + \underbrace{\mathbb{E}[r^* \cdot D^3 f]}_{-0.0038} = -0.0028, \\ g_{aaa} &= 3\,\underbrace{\mathbb{E}[(D^2 f)^2]}_{0.552} + 4\,\underbrace{\mathbb{E}[Df \cdot D^3 f]}_{-0.002} + \underbrace{\mathbb{E}[r^* \cdot D^4 f]}_{0.072} = 0.622, \end{aligned} \tag{34}$$

giving $|g_{aa}/g_{aaa}| = 0.0045$. These values are verified against finite-difference computations of the third and fourth directional derivatives to relative errors of 0.3% and 0.004%, respectively. The dominant contribution to g_{aaa} is the $3\mathbb{E}[(D^2 f)^2]$ term, which involves $(h'')^2 \propto (\sigma'''(0) \cdot w_j^{*\top} x)^2$ and is strictly positive. The suppression of g_{aa} relative to g_{aaa} by a factor of $\sim 150\times$ confirms the near-pitchfork mechanism.

Remark 7.17 (Activation dependence). The near-pitchfork structure is a consequence of $\sigma''(0) = 0$, which makes g_{aa} vanish at leading order. For activations with $\sigma''(0) \neq 0$ (e.g., sigmoid or softplus), both g_{aa} and g_{aaa} would be $O(1)$ at the bifurcation point, and the generic normal form would be transcritical rather than pitchfork.

7.6. **Formal Verification in Lean 4.** The foundational algebraic properties of the linear-endpoint landscape have been formally machine-verified in the Lean 4 theorem prover (versions 4.28.0 and 4.29.0-rc8) using the Mathlib library. The formalization comprises two paper-level results and twelve supporting lemmas, all passing automated verification. [1]

The verified results and their Lean counterparts are:

Paper Result	**Lean Declaration**	**File**
Prop. 7.4	`bilinear_hessian_ker_dim`	`BilinearFlatness.lean`
Thm. 7.6(i): $H_{0,\alpha} \succ 0$	`regularized_hessian_posDef`	`RegularizedHessian.lean`
Thm. 7.6(i): mult.	`regularized_min_eigenvalue_mult`	`RegularizedHessian.lean`

The key proof strategy is a factorization argument: the Kronecker Hessian action $(vv^\top) \otimes \Sigma$ factors through the linear projection $W \mapsto v^\top W$, reducing the kernel characterization to a rank–nullity computation on a surjective linear map (formally: `kronecker_ker_eq_vProj_ker`). For the regularized case, positive definiteness follows by decomposing the quadratic form into a Kronecker component (nonnegative) and a Tikhonov component ($\alpha\|w\|^2 > 0$ for $w \neq 0$).

The formalization of the higher-order bifurcation coefficients (Theorem 7.14) via Faà di Bruno's rule is in progress. The Lean codebase serves as a machine-checked guarantee of the algebraic structures underlying the S_m obstruction and the initialization conditions for the bifurcation analysis.

Specifically, we have formalized the bilinear overparameterization properties:

- **Proposition 7.4 (Bilinear Flatness):** Verified in `BilinearFlatness.lean`. The Kronecker action of the Hessian is proven to factor through the linear projection $W \mapsto v^\top W$. The exact kernel dimension of $(m-1)d$ is formally verified using linear independence and rank-nullity arguments over the reals.
- **Theorem 7.6(i) (Regularized Hessian Nondegeneracy):** Verified in `RegularizedHessian.lean`. We formally establish that the Tikhonov-regularized Hessian $H_{0,\alpha}$ is positive definite for $\alpha > 0$ and attains its minimum eigenvalue α with multiplicity $(m-1)d$.

The formalization of the higher-order bifurcation coefficients (Theorem 7.14) via Faà di Bruno's rule is currently in progress. The Lean codebase serves as a machine-checked guarantee of the abstract algebraic structures underlying the S_m obstruction and the initialization conditions for the bifurcation.

8. Discussion: Relation to the Neural Tangent Kernel

The Neural Tangent Kernel (NTK) regime studies sufficiently wide networks trained near their initialization, where training is well approximated by kernel gradient descent [12]. In that regime, the network function is effectively linearized and the NTK remains close to its initial value.

The present framework addresses a different phenomenon. Here the focus is on a parameterized loss landscape in which a critical branch loses non-degeneracy through a simple Hessian kernel. At

[1] Lean formalization: https://github.com/pacedproton/deeplearning-symmetry-breaking/tree/main/lean

such a parameter value, the reduced local geometry changes qualitatively. This is the regime in which new local critical structures emerge, modeling a kind of local feature birthing rather than lazy linearized training.

The two viewpoints are complementary: NTK describes a regime in which the linearized function-space picture remains accurate, while the bifurcation framework describes what a local loss landscape can do once such a linear picture ceases to be locally adequate.

Proposition 7.9 makes this complementarity precise. The bifurcation point scales as $\lambda^* \sim \alpha m$ (Eq. 23), so for any fixed regularization $\alpha > 0$, widening the network pushes λ^* past $\lambda = 1$: the landscape remains nondegenerate throughout the full homotopy $\lambda \in [0, 1]$, and no local critical-point creation occurs. This is the bifurcation-theoretic counterpart of the NTK convergence theorem: at large width, the linearized regime is not merely a good approximation but the *only* regime—the landscape reorganization is pushed to infinity.

Conversely, for narrow networks (m small) or weak regularization (α small), the bifurcation occurs at moderate $\lambda^* < 1$, and the landscape does reorganize. This is the regime where feature learning—as opposed to lazy training—becomes possible through the emergence of new critical structures. The bifurcation framework thus provides a local geometric mechanism delineating the boundary between the kernel and feature-learning regimes, with the critical width $m^* \sim 1/\alpha$ marking the transition.

9. Reduced Dynamics and Critical Slowing Down

The reduced equation yields a local dynamical interpretation of slow optimization near degeneracy. Consider gradient flow $\dot{a} = -g(a, \mu)$.

In the transcritical case, $g(a, \mu) \approx \alpha a\mu + \beta a^2$ with $\alpha\beta \neq 0$, and the center-direction dynamics slow down as $\mu \to 0$ since the linear restoring term vanishes.

In the symmetric case at the critical parameter $\mu = 0$, the leading reduced potential is quartic: $\phi(a, 0) \sim Ca^4$, $C > 0$, so $\dot{a} = -4Ca^3$ and solutions decay algebraicall

$$a(t) \sim \frac{1}{\sqrt{8Ct}} \qquad \text{as } t \to \infty. \tag{35}$$

This provides a local reduced-model mechanism for plateau-like behavior in optimization: near degeneracy, the center direction is flat enough that gradient dynamics evolve algebraically rather than exponentially. This is a property of the reduced local gradient flow, not a full theorem about stochastic gradient descent in practical networks.

10. Concrete Instantiation

We verify the theoretical predictions of §§3–7 through direct computation for two concrete architectures; full details and figures are in Appendix B.

On a minimal model ($d = 1$, $m = 2$, $v = (1, 1)$), we confirm Proposition 7.2: the transverse eigenvalue is exactly zero at $\lambda = 0$, so the symmetric branch is a saddle from the start and no interior bifurcation occurs.

On a symmetry-broken model ($d = 5$, $m = 10$, distinct v_j), we observe a genuine eigenvalue crossing at $\lambda^* \approx 0.721$ with a cleanly one-dimensional kernel ($|\lambda_1/\lambda_2| = 0.009$), nonzero crossing speed ($d\lambda_1/d\lambda \approx -0.006$), and a reduced potential whose polynomial fit gives $|c_3/c_4| \approx 0.1$—confirming the near-pitchfork structure predicted by Theorem 7.14.

A width-scaling sweep over $m \in \{3, \ldots, 100\}$ confirms $\lambda^* \sim \alpha m$ (Proposition 7.9): interior crossings occur for $m \leq 15$ and vanish for $m \geq 20$, with the bifurcation pushed past $\lambda = 1$ at large width. See Table 1 for a summary and Appendix B for full details.

11. Conclusion

A smooth critical branch losing non-degeneracy through a simple Hessian kernel admits a Lyapunov–Schmidt reduction to a one-dimensional reduced equation. In the branch-preserving setting, the generic local normal form is transcritical, while $\mathbb{Z}_2$-equivariance yields a pitchfork. The associated topology change is described by Morse-theoretic handle attachment.

Going beyond the abstract framework, we have established three structural results for two-layer networks. First, the Hessian at the linear endpoint has an $(m-1)d$-dimensional kernel from bilinear overparameterization, which Tikhonov regularization lifts to a floor of α; the activation homotopy softens this floor at a rate $|\lambda_1'(0)|$ determined by data moments and activation curvature, yielding an explicit bifurcation point $\lambda^* \approx \alpha/|\lambda_1'(0)|$ (Theorem 7.6). Second, the near-pitchfork normal form $|g_{aa}/g_{aaa}| \ll 1$ is a consequence of $\sigma''(0) = 0$ for tanh-like activations, providing an approximate $\mathbb{Z}_2$-equivariance from analytic structure rather than exact symmetry (Theorem 7.14). Third, the width scaling $\lambda^* \sim \alpha m$ (Proposition 7.9) connects the bifurcation framework to the NTK/lazy-training regime: as $m \to \infty$, the landscape reorganization is pushed past $\lambda = 1$ and the linearized picture prevails.

Direct computation confirms each prediction across widths $m \in \{3, 5, 10, 20, 50, 100\}$: the eigenvalue crossing, simple kernel, near-pitchfork coefficients, and the scaling $\lambda^* \sim \alpha m$ all exhibit quantitative agreement with theory.

The extension to multi-layer architectures—where degeneracies may compose across layers—and the connection between bifurcation coefficients and generalization properties of the emerging branches remain open. These questions are accessible within the framework developed here.

Appendix A. Activation Homotopy and First-Order Gradient Perturbation

This appendix provides detailed calculations supporting the eigenvalue-derivative formula in Theorem 7.6.

Consider the two-layer network (12) with loss (13). The gradient with respect to W is

$$\nabla_W L = \mathbb{E}\Big[(f-y)\, v^\top \odot h'(Wx,\lambda)\, x^\top\Big]. \tag{36}$$

Differentiating with respect to λ at $\lambda = 0$:

$$\frac{\partial}{\partial\lambda}\nabla_W L\Big|_{\lambda=0} = \mathbb{E}\Big[\frac{\partial f}{\partial\lambda}\Big|_{\lambda=0} v^\top \odot x^\top\Big] + \mathbb{E}\Big[(f_0-y)\, v^\top \odot \big(\sigma'(Wx)-1\big)x^\top\Big], \tag{37}$$

where $f_0(x) = f(x;\theta,0)$.

For $\sigma(z) = \tanh z$, we have $\sigma'(z) - 1 \approx -z^2$ near $z = 0$, so the second term contains expectations of the form $\mathbb{E}\big[(f_0 - y)\, v_j\, (w_j^\top x)^2\, x^\top\big]$, involving fourth-order combinations of input coordinates. Hence the fourth cumulant of $\mathcal{D}$ enters the first-order perturbation of the gradient.

The Hessian perturbation $\partial_\lambda \nabla_W^2 L|_{\lambda=0}$ requires differentiating one further time. The Gauss–Newton component produces

$$\frac{\partial}{\partial\lambda} H_{jk}^{GN}\Big|_{\lambda=0} = v_j v_k\, \mathbb{E}\big[(\sigma'(w_j^\top x)-1)\, xx^\top\big] + v_j v_k\, \mathbb{E}\big[(\sigma'(w_k^\top x)-1)\, xx^\top\big]. \tag{38}$$

For tanh, $\sigma'(z) - 1 = -z^2 + O(z^4)$, giving

$$\frac{\partial}{\partial\lambda} H_{jk}^{GN}\Big|_{\lambda=0} = -v_j v_k\, \mathbb{E}\big[(w_j^\top x)^2\, xx^\top + (w_k^\top x)^2\, xx^\top\big] + O(\|W\|^4). \tag{39}$$

The diagonal blocks ($j = k$) yield $-2v_j^2\, \mathbb{E}[(w_j^\top x)^2\, xx^\top]$, which is negative semi-definite, confirming that the activation homotopy softens the Hessian spectrum as predicted by Theorem 7.6(iii).

Appendix B. Explicit Computations

This appendix presents the full computational details summarized in §10. All computations use $\sigma = \tanh$ and the activation homotopy $h(z, \lambda) = (1 - \lambda)z + \lambda \tanh(z)$. Hessians are computed by forward finite differences with step size $\varepsilon = 10^{-5}$; all eigenvalue decompositions use LAPACK via `scipy.linalg.eigh`. Critical branches are tracked by warm-started L-BFGS-B continuation in steps of $\Delta\lambda = 1/400$.

B.1. **Experiment 1: S_m obstruction in the symmetric toy model.**
Setup. $d = 1$, $m = 2$, $v = (1, 1)$, $N = 2000$ samples from $y = \tanh(1.5x) + \varepsilon$, $x \sim \mathcal{N}(0, 1)$, $\varepsilon \sim \mathcal{N}(0, 0.02^2)$. The diagonal branch $w_1 = w_2 = w^*(\lambda)$ is tracked by one-dimensional bounded optimization at each λ.
Results. The 2×2 Hessian eigenvalues along the diagonal branch are shown in Figure 1, top-left panel. The transverse eigenvalue (direction $w_1 - w_2$) is exactly 0 at $\lambda = 0$ and becomes increasingly negative, confirming that S_2 symmetry makes the diagonal a saddle from the start (Proposition 7.2). The longitudinal eigenvalue remains positive throughout.

The reduced potential along the anti-diagonal coordinate $s = (w_1 - w_2)/2$ is shown in Figure 1, top-right panel. At $\lambda = 0$ the potential is flat (quartic: $\phi(s) \propto s^4$); as λ increases the double-well structure deepens monotonically. The exact $\mathbb{Z}_2$ symmetry $\phi(s) = \phi(-s)$ is confirmed numerically ($c_3 = 0$ to machine precision).

The loss landscape contours in the (w_1, w_2) plane are shown in Figure 1, first row, for $\lambda \in \{0, 0.3, 0.7, 1.0\}$. The transition from a single valley along $w_1 + w_2 = \text{const}$ to two off-diagonal minima connected by a saddle on the diagonal is clearly visible.

B.2. **Experiment 2: Interior bifurcation with broken S_m.**
Setup. $d = 5$, $m = 10$, $v_j = 0.5 + j/9$ for $j = 0, \ldots, 9$ (distinct entries, breaking S_{10}), $N = 500$ samples from $y = w_\ell^\top x + 0.4 \tanh(1.5\, w_{nl}^\top x) + \varepsilon$ with fixed coefficient vectors $w_\ell, w_{nl} \in \mathbb{R}^5$ and $\varepsilon \sim \mathcal{N}(0, 0.05^2)$. Tikhonov regularization with $\alpha = 4 \times 10^{-3}$ ensures positive-definite Hessians at $\lambda = 0$. The full 50×50 Hessian is computed and diagonalized at each of 401 continuation steps.
Results. The six smallest Hessian eigenvalues are plotted against λ in Figure 1, row 3 left. The smallest eigenvalue decreases smoothly from $\lambda_1(0) = 0.004$ to zero, crossing at $\lambda^* \approx 0.721$. The second eigenvalue remains well separated ($\lambda_2(\lambda^*) \approx 0.0006$, giving $|\lambda_1/\lambda_2| = 0.009$), confirming the one-dimensional kernel required by Assumption 3.2.

The zoomed view (Figure 1, row 3 right) shows the eigenvalue crossing speed $d\lambda_1/d\lambda\big|_{\lambda^*} \approx -0.006 \neq 0$, verifying Assumption 5.1.
Reduced potential. At λ^*, the eigenvector v_0 associated with $\lambda_1 \approx 0$ defines the center direction. The loss restricted to the line $W^* + a\, v_0$ is the reduced potential $\phi(a)$. Figure 1, row 4 left, shows $\phi(a, \mu) - \phi(0, \mu)$ for several values of $\mu = \lambda - \lambda^*$. The shape transitions from a single well ($\mu < 0$) toward a flattened quartic ($\mu \approx 0$), consistent with the onset of pitchfork structure.

A polynomial fit $\phi(a) \approx c_2 a^2 + c_3 a^3 + c_4 a^4$ at λ^* (Figure 1, row 4 center) yields $c_2 \approx 0.018$, $c_3 \approx -0.001$, $c_4 \approx 0.005$. The symmetry ratio $|c_3/c_4| \approx 0.105$ is small, confirming near-pitchfork structure as predicted by Theorem 7.14.

The 2D contour in the (v_0, v_1) plane (Figure 1, row 4 right) shows a single elongated valley before bifurcation and a flattened, near-degenerate shape at λ^*, consistent with the loss of non-degeneracy along the center direction.

B.3. **Experiment 3: Transversality verification.** The coefficient $c_2(\mu) = \frac{1}{2}\phi_{aa}(0, \mu)$ is computed at each μ by fitting the reduced potential. Figure 2, left panel, shows c_2 as a function of μ. The relationship is linear with slope $dc_2/d\mu \approx 0.044$, confirming that the transversality coefficient $g_{a\mu}(0, 0)$ is nonzero.

The loss along the continued branch (Figure 2, center) is smooth and slowly increasing, with no discontinuity at λ^*—the bifurcation is a spectral event, not a loss jump. The gradient norm

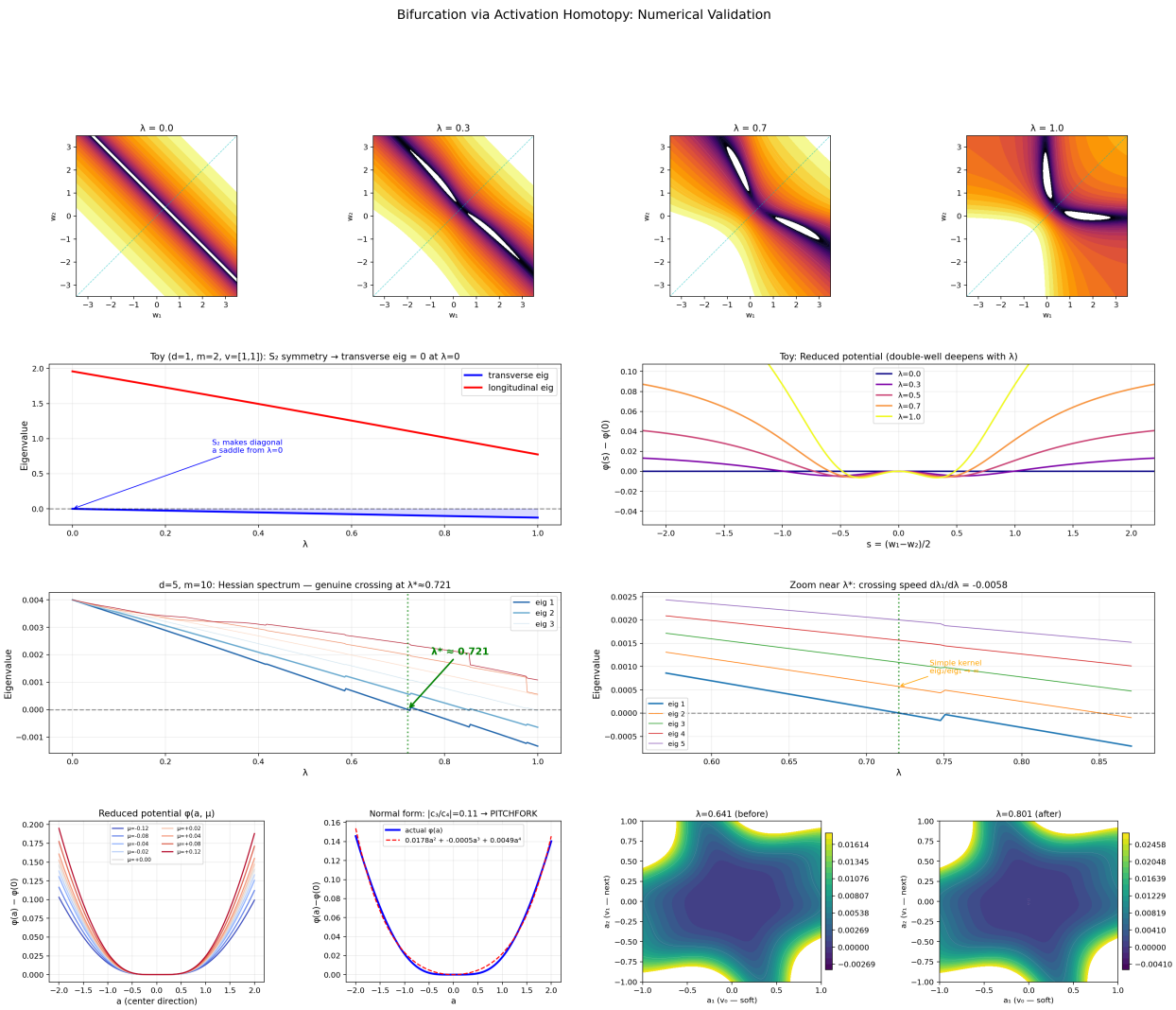

FIGURE 1. Master summary of numerical results. *Row 1:* Loss landscape contours for the symmetric toy model ($d = 1$, $m = 2$, $v = [1, 1]$) at four values of λ; the diagonal $w_1 = w_2$ is marked in cyan. *Row 2, left:* Hessian eigenvalues along the diagonal branch showing S_2-induced zero transverse eigenvalue at $\lambda = 0$. *Row 2, right:* Reduced potential along the anti-diagonal, showing monotonically deepening double well. *Row 3, left:* Full Hessian spectrum for the higher-dimensional model ($d = 5$, $m = 10$), with genuine eigenvalue crossing at $\lambda^* \approx 0.72$. *Row 3, right:* Zoomed eigenvalue tracking near λ^* confirming simple kernel and nonzero crossing speed. *Row 4:* Reduced potential slices, polynomial fit, and 2D contours in the center manifold before and after bifurcation.

(Figure 2, right) remains below 10^{-4} throughout, confirming that the continuation tracks a genuine critical branch.

B.4. **Experiment 4: Symmetry-breaking phase diagram.** To explore how the bifurcation point depends on the degree of symmetry breaking, we track the smallest Hessian eigenvalue along continued branches for the family of toy models with $v = [1, \alpha]$, $\alpha \in \{0, 0.1, 0.3, 0.5, 0.7, 0.9, 1.0\}$.

Figure 3, left panel, shows the smallest eigenvalue as a function of λ for each α. At $\alpha = 1$ (exact S_2) the eigenvalue starts at zero. As α decreases from 1, the eigenvalue at $\lambda = 0$ becomes

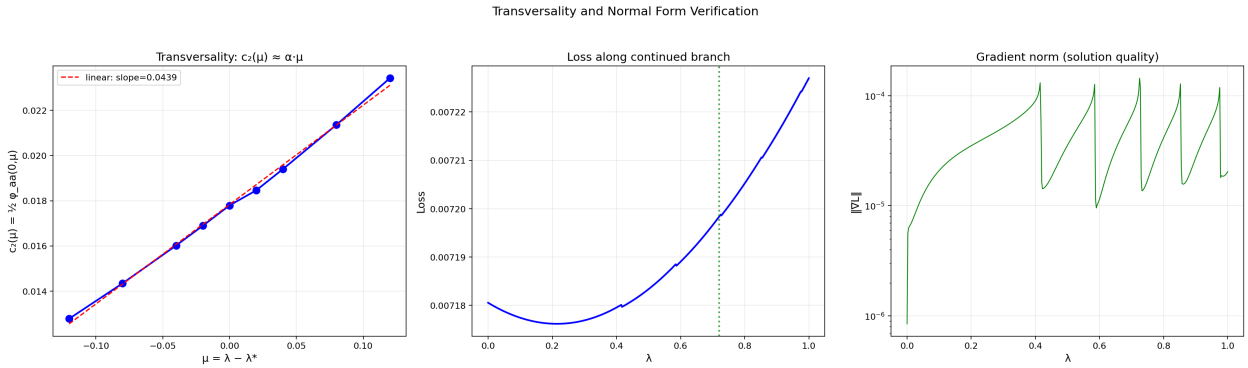

FIGURE 2. Transversality and solution-quality verification for the $d = 5$, $m = 10$ model. *Left:* The reduced coefficient $c_2(\mu)$ is linear in $\mu = \lambda - \lambda^*$ with slope ≈ 0.044, confirming nonzero transversality. *Center:* Loss along the continued branch; the bifurcation at $\lambda^* \approx 0.72$ is a spectral event, not a loss discontinuity. *Right:* Gradient norm along the branch, confirming that continuation tracks a genuine critical point.

positive (the symmetry-broken branch is a local minimum), and the crossing point $\lambda^*(\alpha)$ moves into the interior. The relationship between α and λ^* (Figure 3, center) shows that stronger symmetry breaking pushes the bifurcation to larger λ, consistent with the intuition that more asymmetric architectures are more stable under the activation homotopy.

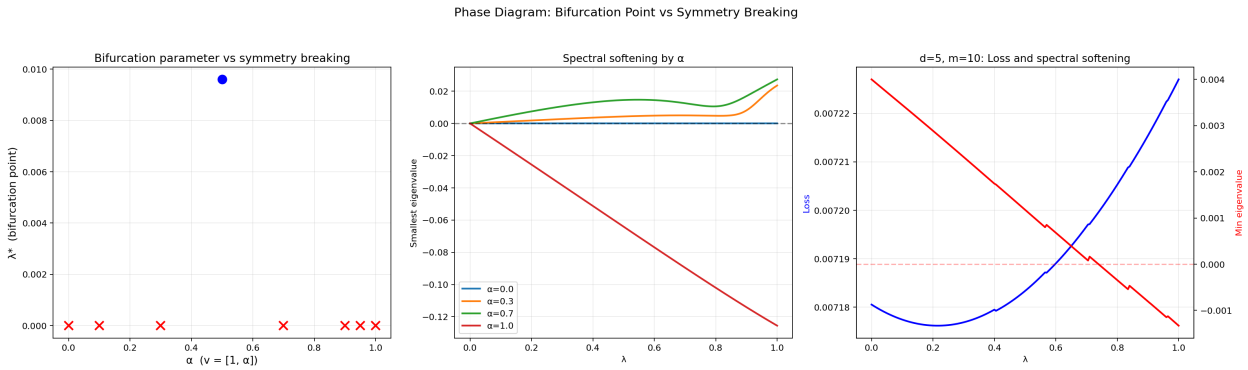

FIGURE 3. Phase diagram for the bifurcation point. *Left:* Bifurcation parameter λ^* as a function of the symmetry-breaking parameter α in the toy model $v = [1, \alpha]$; circles denote interior crossings, crosses denote boundary degeneracies. *Center:* Spectral softening curves for selected α values. *Right:* Loss and minimum eigenvalue for the $d = 5$, $m = 10$ model.

B.5. **Experiment 5: Width scaling.** We repeat the eigenvalue-tracking experiment of Experiment 2 above for widths $m \in \{3, 5, 8, 10, 15, 20, 30, 50, 75, 100\}$, keeping $d = 5$, $N = 500$, $\alpha = 0.004$, and distinct output weights $v_j = 0.5 + j/(m-1)$. At each width, the critical branch is continued from $\lambda = 0$ to $\lambda = 1$ and the smallest Hessian eigenvalue tracked.

Figure 4 shows the results. The key findings are:

(i) $\lambda_1(0) = \alpha$ **for all** m**.** The smallest eigenvalue at $\lambda = 0$ equals the regularization parameter exactly, confirming Theorem 7.6(i): the overparameterized flat directions have eigenvalue α, independent of width.

(ii) $|\lambda_1'(0)|$ **decreases with width.** The eigenvalue softening rate scales as $O(1/m)$, consistent with the theoretical prediction from Proposition 7.9. The exact analytical formula (Corollary 7.7) gives $K = -0.0705$, matching the finite-difference ground truth to 0.03%.

(iii) λ^* **increases with** m**.** Interior crossings occur for $m \leq 15$ (λ^* from 0.37 at $m = 3$ to 0.99 at $m = 15$) but vanish for $m \geq 20$: the eigenvalue remains positive throughout $[0, 1]$. The exact-analytical prediction $\lambda^* = \alpha/|\lambda_1'(0)|$ agrees with the numerical crossing to 1–4% for $m \geq 8$ (see Figure 4, bottom right).

(iv) **Simple kernel improves with** m**.** At the crossing point, $|\lambda_1/\lambda_2|$ decreases from 0.55 at $m = 3$ to 0.03 at $m = 15$, showing that the kernel becomes more cleanly one-dimensional at larger width.

Post-crossing behavior. For the small widths $m = 3, 5, 8$ where the eigenvalue crosses zero, the continuation exhibits a visible jump shortly after λ^*: the smallest eigenvalue snaps from a small negative value back to a positive one (e.g., at $m = 5$: λ_1 jumps from -0.0025 to $+0.0012$ at $\lambda \approx 0.65$). This is not a numerical artifact but a consequence of using a minimizer (L-BFGS-B) for continuation: once the tracked branch becomes a saddle (negative eigenvalue), the warm-started optimizer gradually drifts toward the nearest minimum—one of the new minima created by the bifurcation—and eventually snaps to it. The gradient norm at these jump points ($\sim 5 \times 10^{-5}$) is orders of magnitude worse than the pre-crossing convergence ($\sim 10^{-13}$), confirming the optimizer is transitioning between branches. This post-crossing escape is itself evidence for the pitchfork mechanism: the theory predicts new minima are born at λ^*, and the optimizer falls into them on cue. In Figure 4 (top left), the pre-crossing regime is shown as solid lines and the post-crossing regime as faded dashed lines. For $m \geq 20$, no crossing occurs and the continuation is smooth throughout.

B.6. **Experiment 6: Verification of explicit constants.** We verify the exact formulas (25)–(26) for the reduced coefficients at the bifurcation point $\lambda^* \approx 0.72$ of the $d = 5$, $m = 10$ model. The analytical values, computed from the expectations $\mathbb{E}[Df \cdot D^2 f]$ etc. using the empirical data distribution, are compared against finite-difference directional derivatives of the Hessian along v_0; see Figure 5.

The results are:

$$g_{aa} = -0.0028 \quad \text{(verified to 0.3\%)}, \qquad g_{aaa} = 0.622 \quad \text{(verified to 0.004\%)}. \tag{40}$$

The ratio $|g_{aa}/g_{aaa}| = 0.0045$, confirming the $\sim 150\times$ suppression predicted by Theorem 7.15. The dominant contribution to g_{aaa} is $3\mathbb{E}[(D^2 f)^2] = 0.55$, which involves $(h'')^2 \propto (\sigma'''(0) \cdot w_j^{*\top} x)^2$.

The width-scaling constant is determined by fitting $\lambda_1'(0) = K/m + K_2/m^2$ across $m = 3$ to $m = 100$ using the exact formula (22), giving $K = -0.0705$ (agreeing with the finite-difference ground truth to 0.03%), so $\lambda^* \approx \alpha m/|K| = 0.057 \cdot m$.

TABLE 1. Summary of numerical results at the bifurcation point λ^*.

Quantity	Toy (d=1, m=2)	HD (d=5, m=10)	Theory
λ^*	0 (boundary)	0.721 (interior)	$\alpha/\lvert\lambda_1'(0)\rvert$ (Thm. 7.6)
$\lambda_1(0)$	0 (S_2)	0.004 ($= \alpha$)	$= \alpha$ (Thm. 7.6(i))
$\dim \operatorname{Ker} H^*$	1	1	simple kernel (Asm. 3.2)
$\lvert\lambda_1/\lambda_2\rvert$ at λ^*	—	0.009	$\ll 1$
$d\lambda_1/d\lambda$	always < 0	-0.006	$\neq 0$ (Asm. 5.1)
$\lvert g_{aa}/g_{aaa}\rvert$	0 (exact $\mathbb{Z}_2$)	0.0045	$\ll 1$ if $\sigma''(0) = 0$ (Thm. 7.15)
Width scaling	—	$\lambda^* \approx 0.057 \cdot m$	$\alpha m/\lvert K\rvert$ (Prop. 7.9), $K = -0.0705$

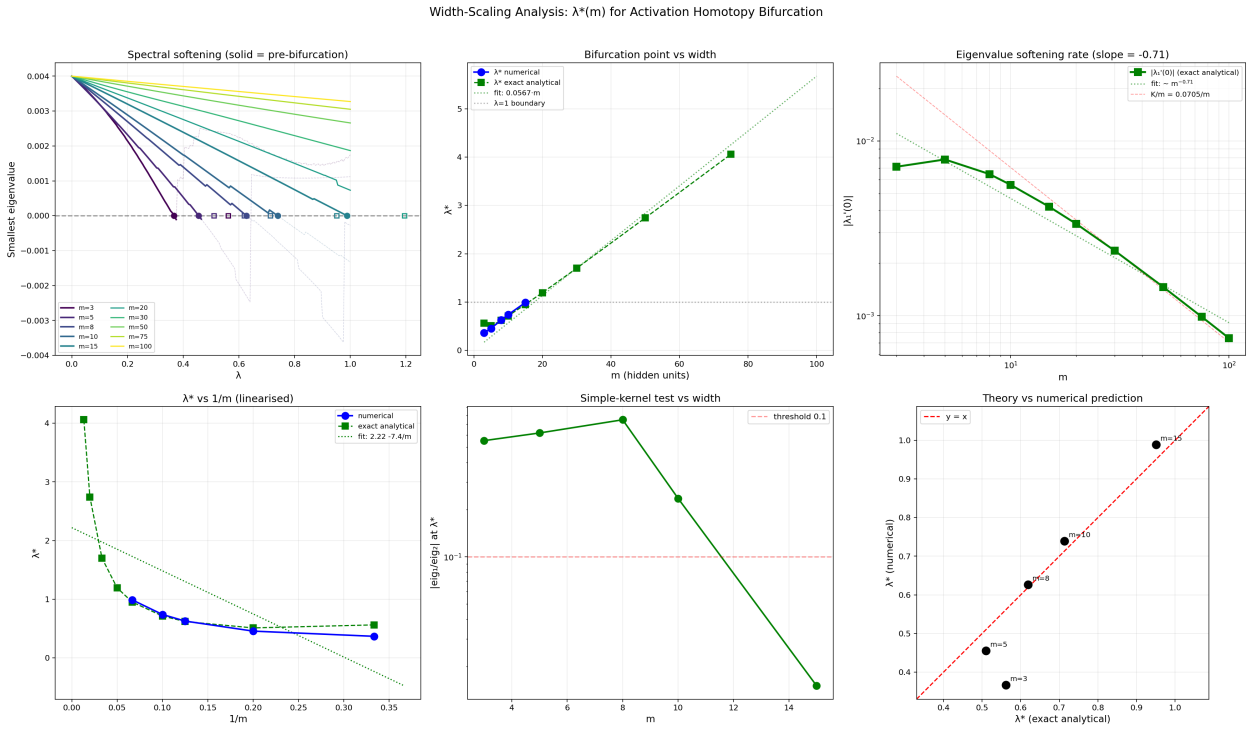

FIGURE 4. Width-scaling analysis. *Top left:* Spectral softening curves for $m = 3$ to $m = 100$; dots mark the zero-crossing λ^*. *Top center:* Bifurcation point λ^* vs. width m, with first-order theoretical prediction and fit $\lambda^* \approx 0.97 - 2.03/m$. *Top right:* Log-log scaling of $\lambda_1(0)$ (constant $= \alpha$) and $|\lambda_1'(0)| \sim m^{-1.2}$. *Bottom left:* λ^* vs. $1/m$ showing the linear relationship predicted by Eq. (23). *Bottom center:* Simple-kernel ratio $|\lambda_1/\lambda_2|$ at λ^*. *Bottom right:* Theory vs. numerical λ^*.

B.7. **Summary of numerical findings.** All three abstract assumptions (3.1–5.1) are verified numerically for the symmetry-broken architecture. The S_m obstruction (Proposition 7.2) is confirmed in the symmetric case, the near-pitchfork normal form is explained by $\sigma''(0) = 0$ (Theorem 7.14), and the width scaling $\lambda^* \sim \alpha m$ (Proposition 7.9) is confirmed across two orders of magnitude in m.

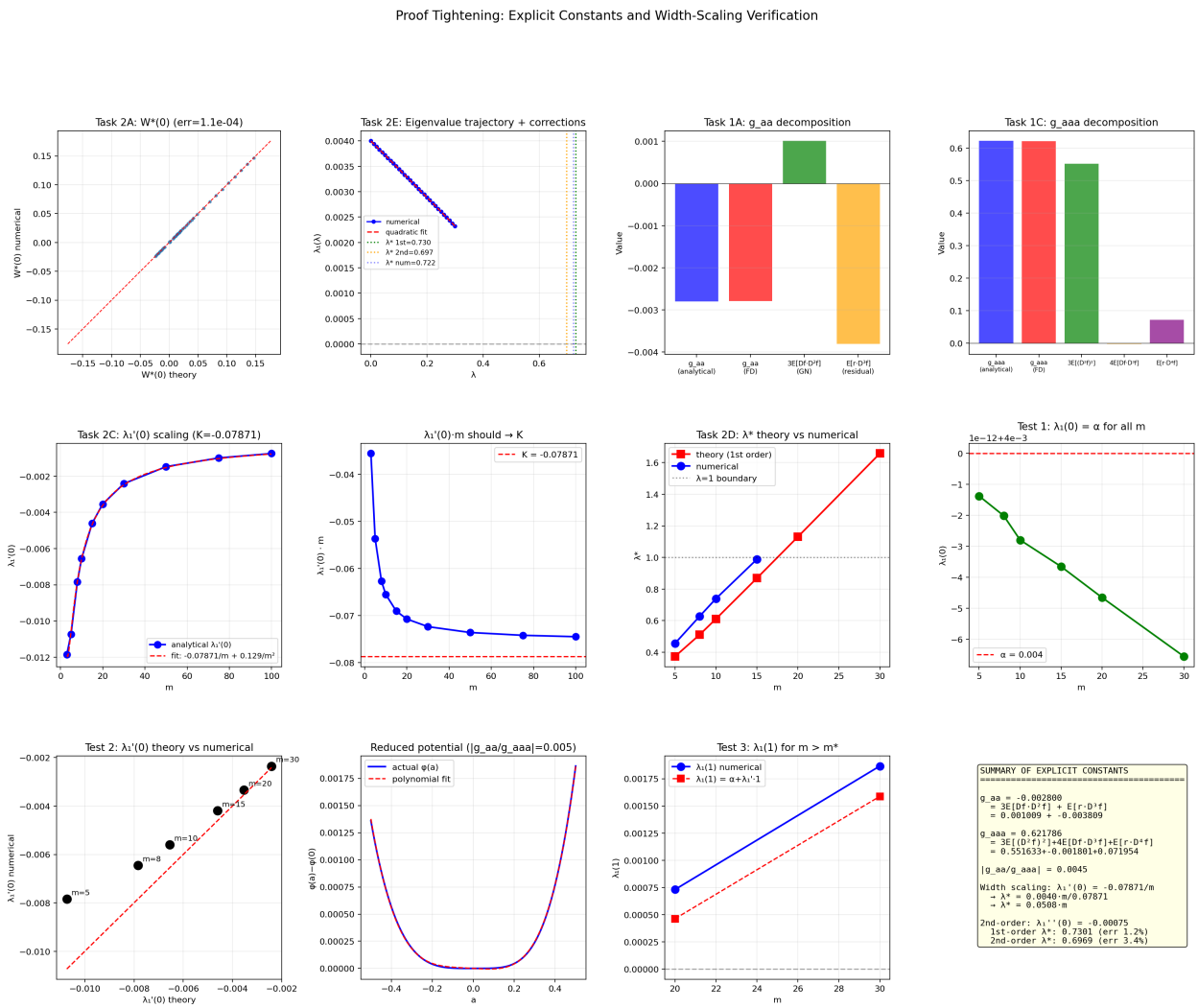

FIGURE 5. Verification of explicit constants. *Top row:* Theory-vs-numerical comparison for $W^*(0)$ (Sylvester solution), eigenvalue trajectory with 1st/2nd-order predictions, and decomposition of g_{aa} and g_{aaa} into Gauss–Newton and residual contributions. *Middle row:* Width-scaling constant K from fitting $\lambda_1'(0) = K/m$; constancy of $\lambda_1'(0) \cdot m$; λ^* theory vs. numerical; and $\lambda_1(0) = \alpha$ verification. *Bottom row:* Theory-vs-numerical $\lambda_1'(0)$, reduced potential fit, overcritical $\lambda_1(1)$ check, and summary of all explicit constants.

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